MorphGrower: A Synchronized Layer-by-layer Growing Approach for Plausible Neuronal Morphology Generation

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01 MorphGrower: Title, Authors, and Context

- Topology-aware, synchronized layer-by-layer CVAE (conditional variational autoencoder).
- Joint sibling-branch generation for **tree consistency**
- Bifurcation-only topology; growth snapshots for interpretability

02 Motivation: Limits of Rule-based Methods and MorphVAE

- Rule-based approaches are slow, sensitive, and cell-type specific
- MorphVAE (variational autoencoder) weakly models inter-branch dependencies and yields invalid multifurcations
- Generation with real-tree references encourages **biologically grounded** growth

03 Related Work Landscape: Rule-based vs MorphVAE vs MorphGrower

- Rule-based methods encode handcrafted priors and enforce **topological validity**
- MorphVAE uses sequence modeling on 3D walks with long decoding issues
- MorphGrower is progressive with sibling-pair CVAE for stronger realism

04 Our Idea and Contributions: Layer-wise Sibling-Pair Generation

- Joint sibling generation captures dependencies and preserves **bifurcations only**
- Layer conditioning on prior growth encodes **developmental priors**
- Demonstrated higher plausibility and functional alignment than prior learning methods

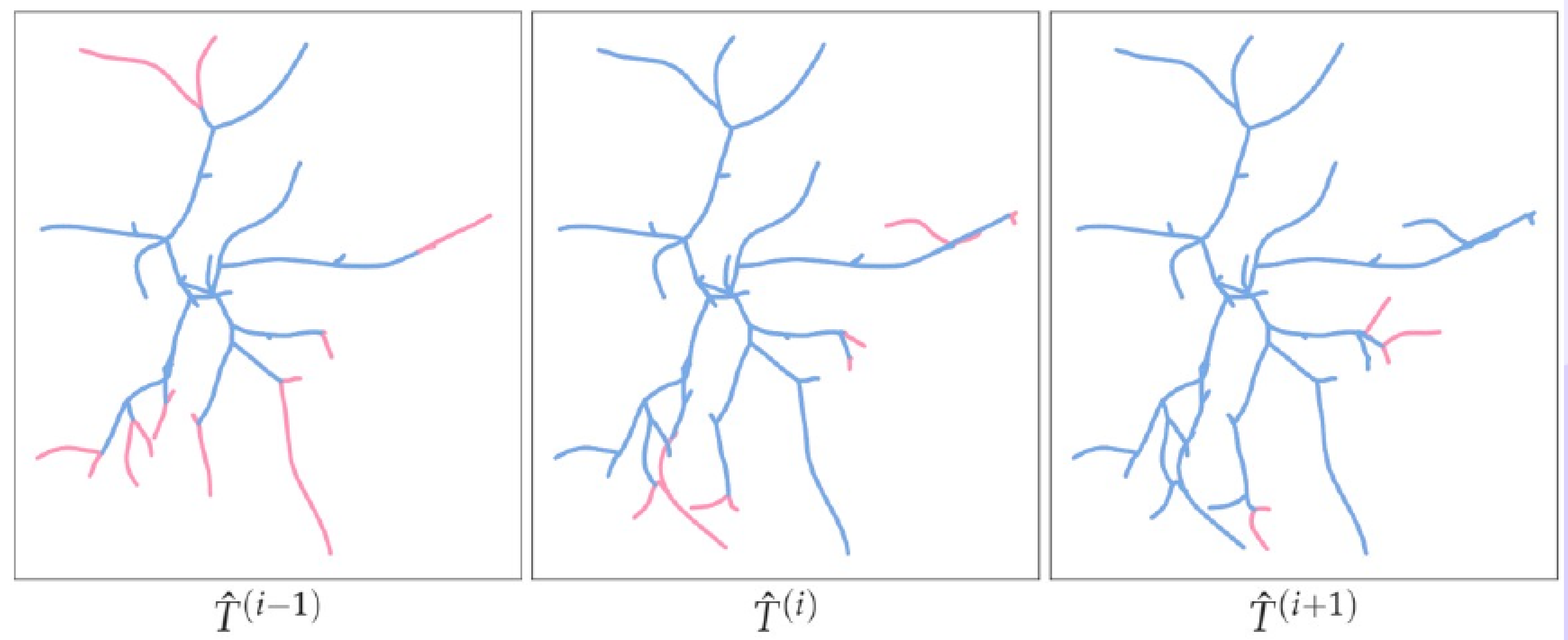
05 Tree Morphology Basics and Branch Layers

- Soma is the root, bifurcations have two children, and tips are leaves with **no offspring**
- Branches are paths between a multifurcation and a downstream tip or bifurcation
- BFS (breadth-first search)-ordered branch layers enable synchronized generation across the tree

06 Method Intuition: Why Sibling-Pair, Layer-wise Generation?

- Layer-wise strategy mirrors biological growth and reduces one-shot complexity
- Sibling pairs exhibit strong dependencies that benefit **joint generation**
- Prior growth informs local ancestors and the global forest for coherent coverage

07 Growth Snapshots: Progressive Layer-wise Generation



- Consecutive plane snapshots highlight new layers in pink for **transparent growth**
- Intermediate trees reveal stable expansion without crossings
- Layer synchronization clarifies sibling-pair decisions at each step

08 Qualitative Growth Micro-cases: Layer i to i+**1** transitions

- Self-avoidance emerges from global conditioning with smooth curvature
  - Siblings bend into nearby gaps; avoid crossings near a crowded bifurcation
  - Paths stay smooth without sharp kinks
- Ancestor-informed orientation yields **balanced angles** and gentle turns
  - Initial tangents follow the parent curve; no sudden departures
  - Both children extend with gradual, symmetric turns
- Coverage balancing allocates siblings toward sparse sectors with even spread
  - One child fills the under-covered sector; the other stays near dense areas
  - Reduces visible holes without overlap

09 Conditioning: Global Forest and Local Ancestors; von Mises–Fisher Latent

- Global context summarizes prior-layer forest with graph aggregation (GNN: graph neural network)
- Local context aggregates ancestor paths with temporal smoothing (LSTM: long short-term memory)
- Directional latent (vMF: von Mises–Fisher) and autoregressive decoder emit sibling branches with stable shapes

10 Generation Protocol: Reference Layers and Soma Handling

- New morphologies grow auto-regressively by reference layer indices with **intermediate trees**
- Soma layer can be provided to stabilize early orientation
- Unconditional soma option exists but may drift global alignment

11 Experimental Setup: Datasets and Baseline

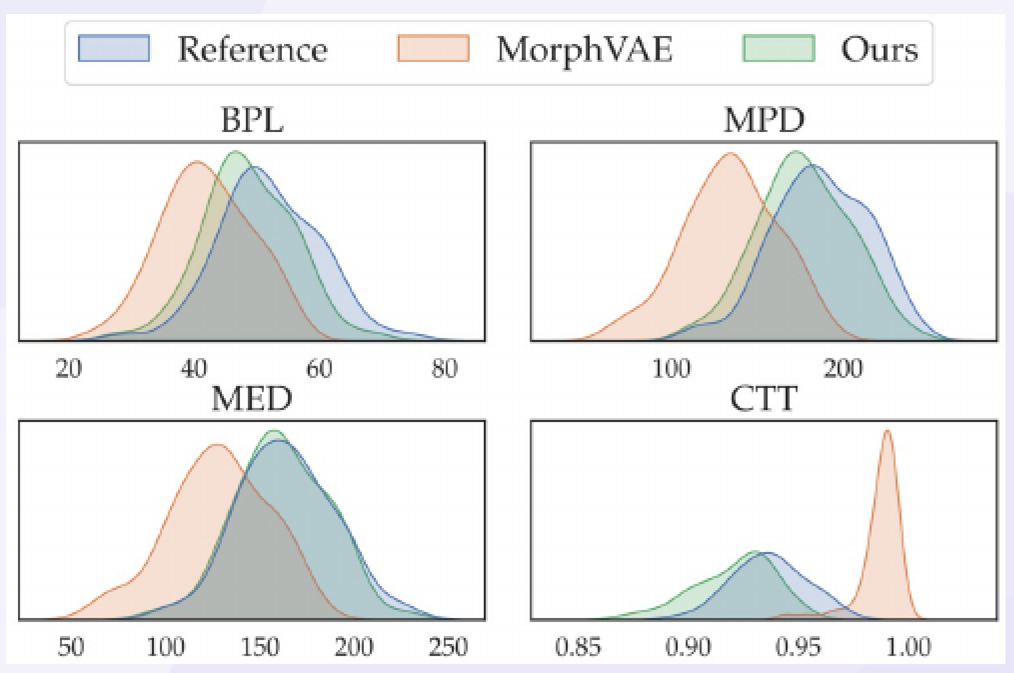
- Four mouse datasets: VPM (ventral posteromedial thalamus), RGC (retinal ganglion cells), M1-EXC (motor-cortex excitatory), M1-INH (motor-cortex inhibitory); **standard splits**
- Baseline is MorphVAE (variational autoencoder) with tuned hyperparameters for **fair comparison**
- Embeddings are matched across methods to ensure comparability

12 Datasets at a Glance (concise)

- VPM and RGC contain hundreds of reconstructions for **robust evaluation**
- M1-EXC includes hundreds while M1-INH includes tens with limited scale
- Cell-type context and sizes guide interpretation of visuals and metrics

13 Metric Distributions: VPM (Qualitative Reading)

- VPM distributions (MPD, BPL, MED, CTT): MorphGrower closely tracks Reference with tighter histograms/CDFs.
- MorphVAE skews longer and appears less compact across these metrics.
- Branch-pair generation and branch-level resampling improve overall shape fidelity.



14 Qualitative Case Studies: RGC and M1-EXC Visual Comparisons

- RGC examples show even sector coverage and self-avoidance with smooth flow
- M1-EXC results retain **balanced angles** and continuous trajectories
- MorphVAE tends toward kinks and branch crowding in deeper layers

15 Real-vs-Fake Classifier Results

Method \ Dataset	**VPM**	**RGC**	**M1-EXC**	**M1-INH**
MorphVAE	86.75 ± 06.87	94.60 ± 01.02	80.72 ± 10.58	91.76 ± 12.14
MorphGrower	54.73 ± 03.36	62.70 ± 05.24	55.00 ± 02.54	54.74 ± 01.63

- Classifier accuracy remains near chance for MorphGrower indicating high realism
- MorphVAE classification remains well above chance across datasets
- Multi-view ResNet-18 CNN (convolutional neural network) blocks feed a linear head for **consistent assessment**

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Whole-Arbor Qualitative: Coverage, Curvature, Crossings

- Even coverage: siblings fill sparse sectors without crowding
- Smooth curvature: no kinks; gradual turns across layers
- Fewer crossings/overlaps: global context steers branches into free space

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Qualitative Recap: What Makes Outputs Look Real?

- Self-avoidance: siblings steer into free space to avoid crossings/overlaps at whole-arbor scale
- Smooth curvature: branches extend with gentle, continuous bends (no kinks)
- Balanced coverage: angles/lengths fill sparse regions without crowding dense sectors

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Quantitative Snapshot: Metric Alignment (brief)

- Across six L-measure metrics MorphGrower outperforms MorphVAE on **most comparisons**
- Branch length and path distance show clear gains with pairing
- Amplitude and orientation metrics remain **close to reference**

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Functional Validation: Electrophysiology Alignment

- Averaged membrane potentials nearly overlap between real and generated with low errors
- Key waveform characteristics align across stimulation phases with **stable peaks**
- Functional similarity corroborates morphological plausibility

20 Most Surprising Finding: Strong realism and functional match with minimal attributes

- Near-chance discrimination and matched electrophysiology occur without diameters
- Compact conditioning suffices for **visual realism** and function
- Minimal attributes still yield robust generalization across datasets

21 Qualitative Panels: M1-INH Axonal Patterns and Coverage

- Collateral spacing appears plausible with broad coverage and smooth bends
- Loops and self-intersections are avoided under global context
- MorphVAE shows zig-zag trajectories and early terminations

22 Failure vs Fix Micro-cases: Conditioning Resolves Conflicts

- Global and local conditioning prevent crossings by steering into free space
- Coverage gaps diminish through balanced sector allocation
- Ancestor alignment mitigates orientation kinks at departures

23 Failure Modes and Limits: When MorphGrower Struggles

- Overly symmetric siblings and local crowding reduce natural variability
- Unconditional soma can drift orientation and deep layers can jitter
- Rare soma-edge motifs remain underrepresented without targeted data

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Ablations Summary (brief, qualitative takeaways)

- Sibling-pair generation, global context, and directional latent each provide clear gains
- Provided soma stabilizes **early layers** and global orientation
- Shorter resampling degrades smoothness with rough trajectories

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Conclusion and Outlook

- MorphGrower delivers plausible topology-aware growth and surpasses MorphVAE
- Future work will add diameters and model **asynchronous dynamics**
- Scalable generation can accelerate studies of **neural computation**

Thank You

Questions & Discussion